

DB-Fusion: Dual-Balanced LiDAR-Camera Fusion for 3D Object Detection

Seungjoon Lee, Taeyeong Kim, Useok Choi, Seungjae Kim, Chanse Oh, and MyeongAh Cho

Abstract—Reliable 3D object detection is a fundamental perception capability for autonomous driving systems operating in complex transportation environments. Recent LiDAR-camera fusion methods project multimodal features into Bird’s Eye View (BEV) space to combine LiDAR geometry with camera semantics. However, we observe that BEV-based fusion inherently suffers from two structural imbalances that can limit reliable object perception. First, Modality–Occupancy Imbalance (MOI) arises because dense image-derived BEV features can dominate sparse but geometrically precise LiDAR features. Second, Semantic–Occupancy Imbalance (SOI) arises because foreground objects occupy only a small portion of BEV space, while indiscriminate aggregation over dominant background regions can dilute class-specific object cues. To address these issues, we propose Dual-Balanced Fusion (DB-Fusion), a unified LiDAR–camera fusion framework that explicitly balances modality occupancy and semantic occupancy in BEV representation. DB-Fusion mitigates MOI through depth-guided pseudo-point sampling and geometry-aware modality weighting, while alleviating SOI through semantic-guided BEV partitioning and class-consistent context aggregation. Extensive experiments on the nuScenes benchmark demonstrate that both MOI and SOI are practical bottlenecks in multimodal BEV perception. DB-Fusion achieves 73.2 mAP on the validation set and 73.6 mAP on the test set with competitive NDS. These results demonstrate that explicitly balancing modality occupancy and semantic occupancy is effective for robust 3D object detection in autonomous driving scenarios.

Index Terms—Autonomous driving, Multi-modal 3D object detection, Bird’s-Eye-View representation.

I. INTRODUCTION

RELIABLE 3D perception is a fundamental capability for autonomous transportation systems, where vehicles must continuously understand surrounding traffic participants and road environments to support safe and efficient driving decisions. In particular, 3D object detection plays a central role in localizing and recognizing vehicles, pedestrians, and other road objects. Since perception errors can propagate to downstream planning and control modules and affect vehicle safety [1], accurate 3D object detection is essential not only for individual vehicles but also for the operational reliability of autonomous transportation systems.

To improve 3D perception in complex driving environments, LiDAR–camera fusion has been widely investigated in autonomous driving. Cameras provide rich texture and semantic information but lack reliable 3D geometry, whereas LiDAR

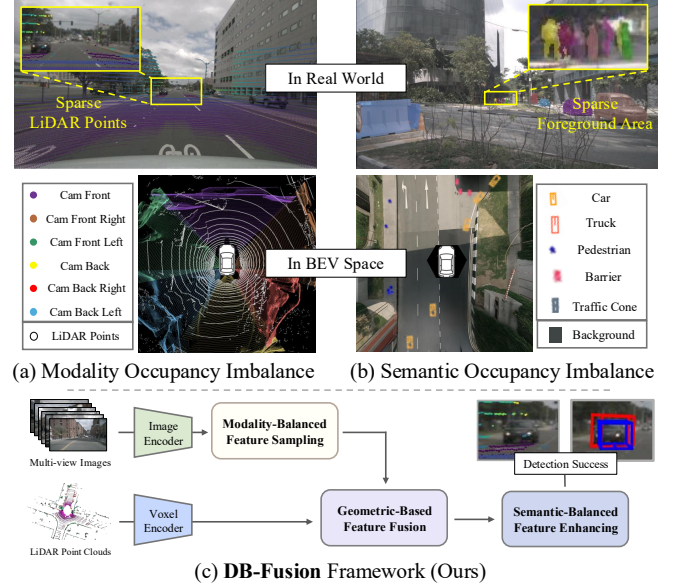


Fig. 1. Illustration of two major imbalance types in LiDAR–Camera fusion. (a) Modality–Occupancy Imbalance (MOI): Top: LiDAR points showing severe long-range sparsity. Bottom: BEV-projected camera features revealing the density gap with LiDAR. (b) Semantic–Occupancy Imbalance (SOI): Top: foreground semantics (e.g., pedestrian, barrier) occupy only a small area. Bottom: BEV view showing background dominance and weakened foreground cues. (c) Dual-Balanced Framework (Ours): Our method mitigates both imbalances via adaptive sampling, geometry-guided fusion, and semantic enhancement.

provides accurate spatial measurements but suffers from point-cloud sparsity, long-range degradation, and occlusion [2], [3]. Therefore, combining image and point-cloud information is necessary to exploit their complementary strengths for reliable 3D object detection. Existing surveys indicate that various fusion paradigms have distinct advantages and limitations, and that representation choice is critical due to the different structures, resolutions, and sensing characteristics of images and point clouds [2]–[4]. Among recent paradigms, BEV-based fusion has become a standard approach for multimodal 3D object detection by projecting both modalities into a unified top-down space [5], [6]. While this paradigm enables natural cross-modal alignment and structured autonomous driving perception, we observe that BEV-space fusion introduces two fundamental yet overlooked structural imbalances that can degrade detection quality and robustness: one between modalities and the other between foreground and background.

As illustrated in Fig. 1(a) and Fig. 1(b), these imbalances take two forms. (i) Modality–Occupancy Imbalance (MOI) arises because LSS-style lifting [7] produces dense image

Seungjoon Lee, Taeyeong Kim, Useok Choi, Seungjae Kim, Chanse Oh, and MyeongAh Cho are with the Department of Software Convergence, Kyung Hee University, Republic of Korea.

(Corresponding author: MyeongAh Cho)

features across every BEV cell, while LiDAR features remain inherently sparse. This density gap allows the sheer volume of camera signals to suppress geometrically precise LiDAR cues, biasing the model toward the dense image branch. (ii) Semantic-Occupancy Imbalance (SOI) stems from the severe spatial disparity between foreground and background in BEV: objects of interest occupy only a small fraction of grid cells, while the surrounding background dominates. Recovering such sparse objects requires gathering class-specific semantic cues, yet existing methods [8]–[10] often aggregate context indiscriminately across the entire scene, letting cross-class interference dilute the class-specific cues—road structure for vehicles, sidewalks for pedestrians. Critically, the two imbalances are not independent but tightly coupled. In LiDAR-sparse regions, dense image cues can provide useful support, but without proper modality balancing, they may entangle weak object cues with dominant background semantics. Indiscriminate scene-level context aggregation then compounds this problem by incorporating irrelevant semantic context rather than selectively reinforcing foreground signals, making class-specific information harder to preserve. Addressing each imbalance separately has inherent limitations; the full benefit is achieved only when both are handled jointly.

Unlike LSS-style lifting [7], projection-based methods [11], [12] avoid modality density mismatch by sampling image features only at LiDAR-occupied locations, but discard valuable camera information in LiDAR-sparse regions (e.g., long range). IS-Fusion [8] partially alleviates the foreground-background disparity through instance-scene interaction, yet its class-agnostic aggregation still introduces cross-class interference. Query-based methods [9], [13] sidestep dense BEV fusion by decoding objects directly from sparse queries, but rely primarily on local context and thus underutilize broader scene information. Overall, prior work operates along a single axis—either reducing modality density mismatch or enhancing foreground emphasis—while the two problems fundamentally co-occur and compound each other.

To address these limitations, we propose Dual-Balanced Fusion (DB-Fusion), a unified framework that jointly mitigates MOI and SOI. For MOI, DB-Fusion introduces Modality-Balanced Feature Sampling (Mo-Bal), which supplements LiDAR-guided image sampling with depth-predicted pseudo-points, and Geometric-based Feature Fusion (GeoFuse), which adaptively weights LiDAR and camera features using distance- and angle-dependent sensor reliability. For SOI, Semantic-Balanced Feature Enhancing (Se-Bal) partitions the BEV map into semantic classes and performs class-consistent context aggregation, allowing each object to exploit semantically relevant surroundings rather than the entire scene.

Extensive experiments on the nuScenes benchmark [14] confirm that both imbalances have a measurable impact on detection performance: starting from a 70.7 mAP baseline, mitigating MOI alone improves mAP by +1.6, mitigating SOI alone by +2.1, and addressing both jointly yields 73.2 mAP with competitive NDS—with the largest gains emerging from their combination. DB-Fusion further demonstrates strong generalization on the nuScenes test set, reaching 73.6 mAP. Detailed analysis, attention visualizations, and failure-case

comparisons further demonstrate that each module effectively targets its intended imbalance. Beyond the resulting accuracy, the central contribution of this work is a principled framework that identifies MOI and SOI as the two structural bottlenecks of BEV-based fusion and resolves both within a single pipeline.

II. RELATED WORKS

A. BEV Space Generation

Generating Bird’s-Eye-View (BEV) representations has become fundamental for unified multimodal perception. LiDAR BEV features are typically generated by voxelizing or pillarizing point clouds and aggregating height-wise geometric information into a top-down feature map [15], [16]. Since LiDAR measurements exist only at observed 3D points, LiDAR BEV features remain spatially sparse. In contrast, camera BEV features are commonly generated by lifting image features into 3D space using predicted depth distributions. LSS [7] pioneered this lifting-based paradigm, and subsequent works [17], [18] further improved depth estimation and supervision. However, lifting-based methods tend to produce dense image BEV features, where most BEV cells become active.

This difference creates a structural density gap between LiDAR and camera BEV features. BEVFusion [5] observed that dense image features can suppress sparser but geometrically accurate LiDAR signals, leading to imbalanced fusion. Projection-based methods instead sample image features only at LiDAR-projected locations [8], [11], [12], avoiding dense image BEV generation but discarding potentially useful camera information in LiDAR-sparse regions. Existing approaches therefore tend to favor either uniformly dense lifting or strictly sparse projection, without explicitly balancing feature density according to local LiDAR coverage or modality reliability.

B. LiDAR–Camera Fusion Models

Multimodal fusion leverages the complementary strengths of LiDAR geometry and camera semantics. Existing fusion models can be broadly categorized according to how they align and exchange multimodal information. We review them with a focus on two challenges: balancing modality-specific feature density and preserving class-specific foreground cues.

1) *LiDAR-Based Fusion*: LiDAR-centric approaches enhance LiDAR-side representations by injecting camera information into point or voxel features. CL3D [19] augments sparse LiDAR features with pseudo-LiDAR points and performs point-guided fusion to associate camera semantics with point features. MENet [20] improves the limited use of dense image information in one-to-one point-pixel mapping through one-to-many mapping around LiDAR points. VoPiFNet [21] extends this direction to voxel-level fusion by constructing voxel-pixel pairs and applying cross-modal attention. However, because camera features are still sampled around LiDAR-observed points or voxels, these methods may underutilize image cues in LiDAR-sparse regions.

2) *BEV-Based Fusion*: Recent methods perform fusion in a unified BEV space to align heterogeneous sensor data. BEV-Fusion [5], [6], DAL [13], and CoreNet [22] adopt LSS [7]-style lifting to construct camera BEV representations and fuse

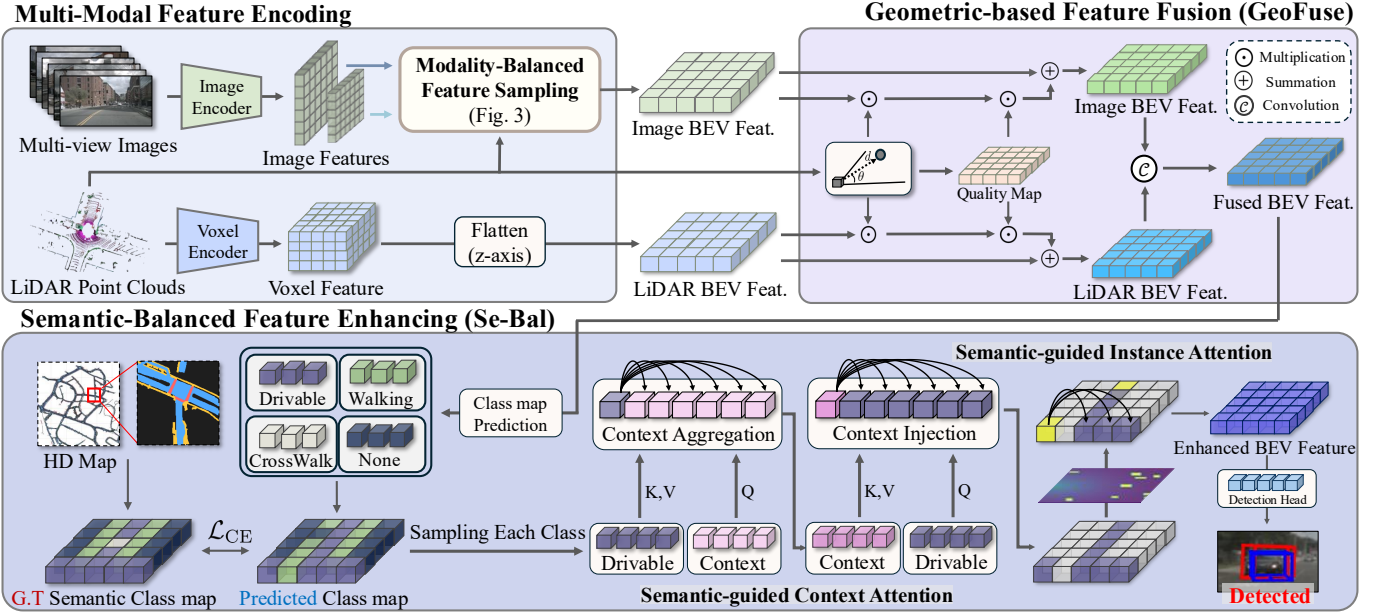


Fig. 2. Overview of DB-Fusion. Multi-view images and LiDAR point clouds are encoded into BEV features. Mo-Bal densifies image BEV features with depth-guided pseudo points, GeoFuse adaptively fuses LiDAR and image BEV features using geometry-derived reliability, and Se-Bal enhances the fused BEV features through semantic-guided contextual aggregation and instance-scene interaction.

them with LiDAR BEV features through concatenation and convolution. These methods achieve effective spatial alignment, but they usually apply similar fusion operations across BEV locations without explicitly considering spatially varying modality occupancy. As a result, dense image-derived BEV features and sparse LiDAR-derived BEV features are fused under the same operation, even though their reliability varies with distance, viewing angle, and local LiDAR coverage. Moreover, because foreground objects occupy only a small fraction of BEV grids, uniform BEV-level aggregation may allow dominant background regions to dilute class-specific foreground cues.

3) *Query-Based Fusion*: Another line of work adopts query-based fusion to exchange multimodal information through sparse object queries or attention mechanisms. Query-based methods [9], [23]–[25] improve efficiency by gathering multimodal features through sparse object-centric interactions. Since these methods avoid direct dense BEV-level association, they can reduce the impact of modality density imbalance. However, their interaction is concentrated around a limited set of queries, which may restrict the use of scene-level context and spatially broad semantic cues. Therefore, query-based methods may not fully exploit scene-wide information.

4) *Refined Fusion Strategies*: In addition to the above fusion paradigms, recent methods have introduced additional strategies to further refine cross-modal interaction. ECL3D [26] enhances LiDAR–camera BEV fusion through depth-semantic feature enhancement and instance-level BEV feature interaction. IS-Fusion [8] introduces instance-scene interaction to enhance object-level representations using surrounding scene context. MambaFusion [10] improves cross-modal exchange through efficient sequence modeling and multi-stage fusion. These strategies improve fusion expressive-

ness from different perspectives, such as feature enhancement, instance-scene interaction, and efficient feature exchange.

However, none of these methods explicitly address the two structural imbalances considered in this work. Although feature enhancement or instance-level aggregation can strengthen foreground representations, they do not explicitly route class-consistent semantic evidence from relevant BEV regions, leaving Semantic–Occupancy Imbalance unresolved. Similarly, refined interaction modules improve information exchange but do not explicitly balance dense image-derived BEV features and sparse LiDAR-derived BEV features. Therefore, existing fusion methods do not jointly resolve Modality–Occupancy Imbalance and Semantic–Occupancy Imbalance within a unified BEV fusion framework.

III. METHOD

A. Overview

As illustrated in Fig. 2, the proposed DB-Fusion framework processes LiDAR point clouds $\mathcal{P}$ and multi-view images $\{I^v\}_{v=1}^{N_{\text{cam}}}$ through modality-specific encoders [15], [27] to obtain LiDAR features F_L and image features F_I . The LiDAR BEV feature $B_L \in \mathbb{R}^{W \times H \times C}$ is obtained by compressing the height dimension of F_L , where W and H denote the spatial dimensions, and C denotes the channel dimension of the BEV grid.

For the image modality, we first apply LiDAR-to-camera projection [8] to obtain anchor image BEV features B_{anchor} at LiDAR point locations. To supplement sparse LiDAR coverage, Modality-Balanced Feature Sampling (Mo-Bal, Section III-B) estimates image-based depth, generates pseudo points at plausible locations, and samples additional camera features to obtain augmented image BEV features B_{aug} . The

final image BEV feature B_I is obtained by fusing B_{anchor} and B_{aug} through a convolutional layer.

Next, Geometric-based Feature Fusion (GeoFuse, Section III-C) models modality reliability based on geometric context, including range and orientation, and learns grid-wise fusion weights to adaptively enhance each modality's features. This produces refined features $\hat{B}_L$ and $\hat{B}_I$, which are concatenated and fused through a convolutional layer to obtain the fused BEV representation B_F .

Subsequently, Semantic-Balanced Feature Enhancing (Se-Bal, Section III-D) partitions the BEV grid B_F into semantic classes using an auxiliary segmentation head, aggregates contextual information within each class, and performs class-wise instance-scene interaction. This produces the final enhanced BEV representation $\hat{B}_F \in \mathbb{R}^{W \times H \times C}$.

The enhanced BEV features $\hat{B}_F$ are fed into a transformer-based decoder [28] that predicts final 3D bounding boxes. Following standard practice, we apply the Hungarian algorithm [29] for matching predictions to ground truth during training. The framework is trained end-to-end with Focal loss [30] for classification, L1 loss for bounding box regression, and auxiliary losses including Smooth L1 loss [31] for depth prediction in Mo-Bal and Cross-Entropy loss for BEV segmentation in Se-Bal.

B. Modality-Balanced Feature Sampling

The core idea of Mo-Bal is to enrich sparse LiDAR-guided sampling with informative image features while preventing dense image features from overwhelming sparse LiDAR signals. As illustrated in Fig. 3, given LiDAR point cloud $\mathcal{P}$ and image features F_I , Mo-Bal constructs an image BEV feature B_I by balancing reliable geometric information with dense semantic cues:

$$B_I = f_{\text{fuse}}(B_{\text{anchor}}(\mathcal{P}), B_{\text{aug}}(\mathcal{P}_{\text{aug}})), \quad (1)$$

where B_{anchor} denotes image BEV features sampled at original LiDAR locations $\mathcal{P}$, B_{aug} represents augmented BEV features sampled at pseudo point locations $\mathcal{P}_{\text{aug}}$, and $f_{\text{fuse}}(\cdot)$ is the depth-guided gating fusion function. The key challenge is to generate $\mathcal{P}_{\text{aug}}$ that maintains cross-modal consistency while supplementing sparse LiDAR coverage.

1) *Depth Prediction With Sparse Supervision*: To generate $\mathcal{P}_{\text{aug}}$, Mo-Bal employs depth estimation to identify geometrically plausible 3D locations. A lightweight depth prediction network $f_{\text{depth}}(\cdot)$ takes image features F_I^v as input and outputs per-pixel depth for each camera view v :

$$\hat{D}^v = f_{\text{depth}}(F_I^v) \in \mathbb{R}^{H_{\text{img}} \times W_{\text{img}}}. \quad (2)$$

The network is trained with sparse LiDAR depth supervision using Smooth L1 loss, producing depth predictions $\{\hat{D}^v\}_{v=1}^{N_{\text{cam}}}$ across all camera views.

2) *Occlusion-Aware Gating*: Using the predicted depth maps $\{\hat{D}^v\}_{v=1}^{N_{\text{cam}}}$, Mo-Bal determines which empty BEV cells are suitable for pseudo point generation. For each empty BEV cell at grid location (x, y) , we generate a candidate point $p_{\text{cand}} = (x, y, z_0)$ at ground height z_0 and project it to all camera views. The projection to view v yields a pixel location

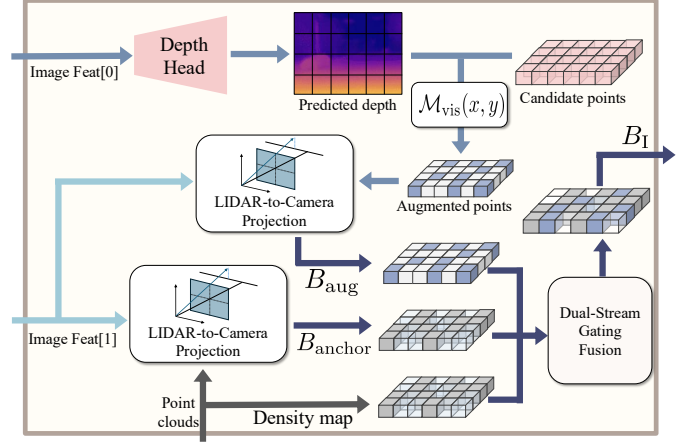


Fig. 3. Illustration of Mo-Bal. Depth prediction identifies visible regions, enabling pseudo point generation at unoccluded areas. Dual-stream gating fusion adaptively combines anchor and augmented BEV features based on LiDAR density.

u^v and the ground-truth depth d_{cam}^v , which is the depth of p_{cand} in the coordinate system of camera v . We then compare d_{cam}^v with the network-predicted depth $\hat{D}^v(u^v)$ at pixel u^v . A cell is marked as visible if the absolute difference is within an adaptive threshold for at least one view:

$$\mathcal{M}_{\text{vis}}(x, y) = \begin{cases} 1, & \text{if } V(x, y) \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where

$$V(x, y) = \{v : |\hat{D}^v(u^v) - d_{\text{cam}}^v| \leq \tau_{\text{abs}} + \tau_{\alpha} \cdot d_{\text{cam}}^v\}. \quad (4)$$

The distance-adaptive threshold $\tau_{\text{abs}} + \tau_{\alpha} \cdot d_{\text{cam}}^v$ accounts for increasing depth uncertainty at longer ranges.

Based on $\mathcal{M}_{\text{vis}}$, Mo-Bal generates pseudo points only at visible locations. Each pseudo point is sampled around the cell center using truncated normal distributions in the xy plane, clipped within cell bounds, and a normal distribution centered at z_0 along the z axis. This produces the augmented point set $\mathcal{P}_{\text{aug}}$.

3) *Dual-Stream Gating Fusion*: Following [8], we obtain $B_{\text{anchor}} \in \mathbb{R}^{W \times H \times C}$ and $B_{\text{aug}} \in \mathbb{R}^{W \times H \times C}$ by sampling image features at $\mathcal{P}$ and $\mathcal{P}_{\text{aug}}$, respectively. The fusion function $f_{\text{fuse}}(\cdot)$ in (1) is realized as $f_{\text{fuse}} = f_{\text{refine}} \circ f_{\text{gate}}$, where f_{gate} computes adaptive weights and f_{refine} produces the final fused features.

Specifically, we first construct a LiDAR density map $D_L \in [0, 1]^{W \times H}$ from the original point cloud $\mathcal{P}$ by normalizing point counts per BEV cell. The fusion is then performed as:

$$G = \sigma(f_{\text{gate}}([B_{\text{anchor}}, B_{\text{aug}}, D_L])), \quad (5)$$

$$B_I = f_{\text{refine}}(G \odot B_{\text{aug}} + (1 - G) \odot B_{\text{anchor}}), \quad (6)$$

where $[\cdot, \cdot, \cdot]$ denotes concatenation, $f_{\text{gate}}(\cdot)$ is a convolutional layer producing the gating map $G \in [0, 1]^{W \times H \times 1}$, $f_{\text{refine}}(\cdot)$ consists of convolutional layers with batch normalization and ReLU activations, and $\odot$ denotes element-wise multiplication.

C. Geometric-Based Feature Fusion

Given B_L and B_I from the previous stage, a straightforward approach is to fuse them directly through convolutional networks to obtain the fused BEV feature B_F . However, this ignores the distinct geometric characteristics of each sensor. LiDAR exhibits high accuracy at close range but degrades at longer distances [16], [32] and varying elevation angles [33]–[35] due to sparsity and measurement uncertainty. In contrast, cameras provide consistent semantic information across different distances and viewing angles. By explicitly accounting for these complementary characteristics, GeoFuse computes distance-dependent sensor reliability and angle-based observation quality from geometric cues, generating confidence maps that adaptively enhance B_L and B_I before fusion.

1) *Distance-Based Reliability Estimation*: For each LiDAR point $p_i = (x_i, y_i, z_i)$, we compute the normalized distance $\hat{d}_i = d_i / \max(d_i)$, where $d_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$. Sensor-specific reliability weights are calculated as:

$$w_L(i) = \sigma(-\alpha \cdot (\hat{d}_i - \beta)), \quad w_I(i) = 1 - w_L(i), \quad (7)$$

where $\sigma(\cdot)$ is the sigmoid function, and α and β are learnable parameters controlling the transition slope and center. This complementary weighting satisfies $w_L + w_I = 1$, reflecting that LiDAR reliability decreases while camera reliability increases with distance.

2) *Angle-Based Quality Modeling*: LiDAR reflection quality degrades at steep elevation angles. For each point, we compute the elevation angle as:

$$\theta_i = \arctan\left(\frac{|z_i|}{\sqrt{x_i^2 + y_i^2}}\right), \quad (8)$$

and evaluate observation quality as:

$$q(i) = \exp\left(-\frac{\theta_i}{\gamma \cdot (1 + 0.5 \cdot \hat{d}_i)}\right), \quad (9)$$

where γ is a learnable parameter that controls the angular decay rate. Points near the horizontal plane, where $\theta_i \approx 0$, receive higher quality scores, while the distance-adaptive term $(1 + 0.5 \cdot \hat{d}_i)$ relaxes the constraint for nearby objects where steep angles are more common.

3) *BEV Mapping and Feature Refinement*: The point-wise reliability weights $w_L(i)$ and $w_I(i)$ and angle quality scores $q(i)$ are mapped to BEV grids by converting points to grid coordinates and averaging values from multiple points per cell. This produces BEV confidence maps M_L , M_I , and $M_{\text{quality}} \in \mathbb{R}^{W \times H}$. Using these maps, we enhance each modality's features:

$$\hat{B}_m = B_m \odot (1 + M_{\text{quality}} \odot M_m), \quad m \in \{L, I\}. \quad (10)$$

The enhanced features $\hat{B}_L$ and $\hat{B}_I$ are then fused via a 3×3 convolution to obtain B_F , adaptively leveraging each sensor's strengths according to distance and observation geometry.

D. Semantic-Balanced Feature Enhancing

With the fused BEV feature B_F , global attention is commonly used for context aggregation. However, foreground objects have distinct class identities, such as vehicles and pedestrians, and benefit from interacting with semantically relevant context rather than the entire scene indiscriminately. For instance, vehicle detection benefits from road structure context, while pedestrian detection relies on sidewalk and crosswalk cues. To capture such semantic structure, we construct a semantic BEV map by rasterizing the nuScenes high-definition map [14] around the ego pose into a BEV grid aligned with B_F , where map elements are grouped into semantic classes. Se-Bal leverages this semantic partitioning by aggregating context within each class independently, enabling foreground objects to interact with semantically relevant regions.

1) *Semantic Routing via BEV Segmentation*: To partition the BEV grid into semantic classes, we employ an auxiliary segmentation head $f_{\text{seg}}(\cdot)$ that predicts class probabilities over $\mathcal{S} = \{\text{drivable, walkway, crosswalk, none}\}$:

$$P = f_{\text{seg}}(B_F) \in [0, 1]^{|\mathcal{S}| \times W \times H}. \quad (11)$$

Each BEV cell is assigned to the class with maximum probability in P , producing binary masks $\mathcal{M}_s \in \{0, 1\}^{W \times H}$ for each class $s \in \mathcal{S}$. These masks partition the BEV feature into class-specific features:

$$B_s = B_F \odot \mathcal{M}_s \in \mathbb{R}^{W \times H \times C}, \quad s \in \mathcal{S}. \quad (12)$$

This four-class discretization explicitly models road-related regions where objects predominantly appear, with road markings such as lane dividers and stop lines grouped into the drivable class.

2) *Semantic-Guided Context Attention*: To aggregate context within each semantic class while avoiding cross-class interference, Se-Bal compresses class-specific features into compact context vectors to extract shared information and reduce noise, and then injects them back only to the corresponding regions.

For each class $s \in \mathcal{S}$, we flatten the class-specific BEV feature B_s into tokens $X_s = \text{Flatten}(B_s) \in \mathbb{R}^{(WH) \times C}$ and introduce a query bank $Q_s \in \mathbb{R}^{N \times C}$, which is orthogonally initialized. Class-wise context is aggregated via multi-head cross-attention:

$$\tilde{Q}_s = f_{\text{MCA}}(Q_s, X_s, X_s) \in \mathbb{R}^{N \times C}, \quad (13)$$

where $f_{\text{MCA}}(Q, K, V)$ denotes multi-head cross-attention with query Q , key K , and value V . For the crosswalk class, which lies at the boundary between roads and sidewalks, we aggregate context from both drivable and walkway regions and blend them with a learnable weight $\alpha \in [0, 1]$ to capture mixed semantic cues.

The aggregated context $\tilde{Q}_s$ is injected back only to the corresponding class region:

$$\tilde{X}_s = f_{\text{MCA}}(X_s, \tilde{Q}_s, \tilde{Q}_s) \in \mathbb{R}^{(WH) \times C}. \quad (14)$$

After reshaping $\tilde{X}_s$ back to spatial dimensions, we update the class feature with residual gating. This process is applied to

TABLE I

COMPARISON WITH RECENT LiDAR-CAMERA FUSION METHODS ON THE nuScenes BENCHMARK. ALL METHODS USE BOTH LiDAR AND CAMERA MODALITIES, AND THE COMPARED FUSION FRAMEWORKS ADOPT VoxelNet [15] AS THE LiDAR BACKBONE.

Methods	Modality	Camera Backbone	Validation Set		Test Set		FPS
			mAP	NDS	mAP	NDS	
PointPainting [12]	L+C	—	65.8	69.6	—	—	—
PointAugmenting [36]	L+C	ResNet-101 [37]	—	—	66.8	71.0	—
MVP [38]	L+C	DLA-34 [39]	66.1	70.0	66.4	70.5	—
TransFusion [9]	L+C	ResNet-50 [37]	67.5	71.3	68.9	71.6	—
BEVFusion (PKU) [6]	L+C	Dual-Swin-T [40]	67.9	71.0	69.2	71.8	0.8
DeepInteraction [41]	L+C	ResNet-50 [37]	69.9	72.6	70.8	73.4	2.6
BEVFusion (MIT) [5]	L+C	Swin-T [27]	68.5	71.4	70.2	72.9	4.2
CMT [42]	L+C	ResNet-50 [37]	70.3	72.9	72.0	74.1	2.8
SparseFusion [23]	L+C	ResNet-50 [37]	70.4	72.8	72.0	73.8	5.3
ObjectFusion [43]	L+C	Swin-T [27]	69.8	72.3	71.0	73.3	—
UniTR [24]	L+C	Swin-T [27]	70.5	73.3	70.9	74.5	4.9
DAL [13]	L+C	ResNet-50 [37]	71.5	74.0	72.0	74.8	4.3
SparseLIF [25]	L+C	V2-99 [44]	71.2	74.6	—	—	2.9
IS-Fusion [8]	L+C	Swin-T [27]	72.8	74.0	73.0	75.2	3.2
ECL3D [26]	L+C	ResNet-50 [37]	72.6	74.8	72.8	75.2	5.6
MambaFusion [10]	L+C	Swin-T [27]	72.7	75.0	73.2	75.9	4.7
CoreNet [22]	L+C	Swin-T [27]	72.5	74.6	73.3	75.6	4.5
SMDFFusion [45]	L+C	Swin-T [27]	72.1	74.5	71.9	74.6	8.4
DB-Fusion (Ours)	L+C	Swin-T [27]	73.2	74.5	73.6	75.0	3.5

all classes except none. Finally, the enhanced class-specific features are merged:

$$\tilde{B}_F = \sum_{s \in S} B_s \in \mathbb{R}^{W \times H \times C}. \quad (15)$$

3) *Semantic-Guided Instance Attention*: To further enhance the BEV feature with instance-level information, we first extract instance candidates from $\tilde{B}_F$. A heatmap prediction head generates a spatial confidence map, from which we select the top- K peak coordinates and gather their features to form an instance token set $I \in \mathbb{R}^{K \times C}$. Each instance token is assigned to a semantic class based on the maximum probability at its location. Tokens within the same class are grouped, and multi-head self-attention is applied within each group:

$$\tilde{I}_s = f_{\text{MSA}}(I_s), \quad s \in \{\text{drivable, walkway, crosswalk}\}. \quad (16)$$

This allows same-class instances to cross-reference one another, producing refined instance representations. These refined tokens are then injected back to their corresponding class regions via cross-attention, where the class-specific BEV feature B_s serves as queries:

$$B_s \leftarrow B_s + \eta_s \cdot (f_{\text{MCA}}(B_s, \tilde{I}_s, \tilde{I}_s) \odot \mathcal{M}_s), \quad (17)$$

where $\eta_s \in [0, 1]$ is a learnable gating parameter, and $\mathcal{M}_s$ spatially restricts the update to the corresponding class region. After updating all class-specific features, we reconstruct the final enhanced BEV feature:

$$\hat{B}_F = \sum_{s \in S} B_s \in \mathbb{R}^{W \times H \times C}. \quad (18)$$

This final BEV feature $\hat{B}_F$ is then fed into the detection decoder to produce 3D bounding box predictions, benefiting from both class-consistent context aggregation and instance-scene interaction that sharpen object evidence while suppressing irrelevant background interference.

IV. EXPERIMENTS

A. Experimental Setup

1) *Dataset and Evaluation*: We evaluate DB-Fusion on the nuScenes dataset [14], a large-scale benchmark for multimodal 3-D object detection in autonomous driving. The dataset contains 1000 driving scenes collected in Boston and Singapore, where each scene provides a full 360-degree field of view using six surround-view cameras and one 32-beam LiDAR sensor. For robustness analysis, we further evaluate the models on nuScenes-C [46], which applies camera-only corruptions at five severity levels while keeping LiDAR inputs unchanged. We report mean Average Precision (mAP) and the nuScenes Detection Score (NDS), where NDS jointly considers detection accuracy and true-positive quality terms, including translation, scale, orientation, velocity, and attribute errors. Unless otherwise specified, all experiments are conducted on the nuScenes validation set.

2) *Implementation Details*: We use a Swin Transformer pretrained on nuImages as the image encoder and VoxelNet [15] as the LiDAR voxel encoder. For detection, we adopt the TransFusion-L head [9], which also serves as our LiDAR-only baseline. To establish a LiDAR-camera fusion baseline, we incorporate the Point-to-Grid (P2G) module from IS-Fusion [8] into TransFusion-L and denote the resulting model as Baseline-LC. Following the standard nuScenes setting, we use a voxel size of $[0.075, 0.075, 0.2]$ and a point cloud range covering a $108 \text{ m} \times 108 \text{ m}$ BEV area, yielding a 180×180 BEV feature map. Input images are resized to 384×1056 , and the Swin encoder is kept frozen during training. In Se-Bal, the segmentation head f_{seg} is supervised using a cross-entropy loss. For class-consistent instance-scene interaction, we retain the top-200 proposals for each frame. All models are trained on eight RTX 3090 GPUs with a batch size of two per GPU.

TABLE II
FULL MOI ANALYSIS ON nuScenes-C [46] UNDER THREE CAMERA-ONLY CORRUPTION TYPES ACROSS FIVE SEVERITY LEVELS.

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion [8]	70.16	72.44	68.40	71.43	65.69	69.90	62.84	68.40	61.32	67.66	65.68	69.99
MambaFusion [10]	69.91	73.40	67.78	72.19	65.20	70.74	64.46	70.41	64.51	70.46	66.37	71.44
DB-Fusion (Ours)	71.15	73.46	69.79	72.74	68.16	71.79	66.60	71.02	65.48	70.45	68.24	71.89

(a) Gaussian corruption

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion [8]	71.53	73.18	70.61	72.70	68.96	71.77	66.65	70.46	63.69	68.85	68.29	71.39
MambaFusion [10]	71.33	74.23	70.35	73.67	68.62	72.65	66.07	71.20	64.55	70.46	68.18	72.44
DB-Fusion (Ours)	72.27	74.05	71.51	73.62	70.24	72.97	68.83	72.20	67.28	71.38	70.03	72.84

(b) Uniform corruption

Method	Severity 1		Severity 2		Severity 3		Severity 4		Severity 5		Average	
	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS	mAP	NDS
IS-Fusion [8]	69.82	72.27	67.68	71.04	66.02	70.09	63.23	68.59	61.61	67.75	65.67	69.95
MambaFusion [10]	69.09	72.90	67.06	71.73	65.39	70.85	64.30	70.32	64.42	70.40	66.05	71.24
DB-Fusion (Ours)	70.84	73.32	69.48	72.54	68.53	72.00	66.70	71.04	65.71	70.56	68.25	71.89

(c) Impulse corruption

TABLE III
FALSE-NEGATIVE ANALYSIS SHOWING THE EFFECT OF SOI.

Method	$\tau = 0.5 \downarrow$	$\tau = 1.0 \downarrow$	$\tau = 2.0 \downarrow$	$\tau = 4.0 \downarrow$	Range	DB-Fusion	IS-Fusion [8]
DB-Fusion (Ours)	15,818	8,305	5,056	3,722	~ 10 m	572	622 (+50)
IS-Fusion [8]	16,235 (+417)	8,496 (+191)	5,063 (+7)	3,727 (+5)	~ 20 m	2,597	2,634 (+37)
MambaFusion [10]	16,060 (+242)	8,664 (+359)	5,566 (+510)	4,294 (+572)	~ 30 m	4,226	4,318 (+92)
					~ 40 m	4,247	4,313 (+66)
					~ 50 m	4,176	4,348 (+172)

(a) FN under different matching thresholds (τ).

(b) FN by range at $\tau = 0.5$ m.

B. Benchmark Results

As shown in Table I, DB-Fusion achieves 73.2 mAP on the nuScenes validation set and 73.6 mAP on the test set, placing it among the top-performing LiDAR-camera fusion methods while maintaining an inference speed of 3.5 FPS, which is comparable to other Swin-T-based approaches. The improvement of DB-Fusion is more pronounced in mAP than in NDS. This is because DB-Fusion primarily recovers objects that previous methods miss entirely, rather than only refining the localization quality of already detected objects. Newly recovered detections, especially those at long range or from rare classes, may still have relatively less precise orientation and velocity estimates, which limits the improvement in the true-positive quality terms included in NDS. This pattern is consistent with our design goal of reducing false negatives caused by MOI and SOI, and we provide further supporting evidence in Sections IV-C and IV-D.

C. Analysis of Imbalance Bottlenecks

1) *Analysis of Modality-Occupancy Imbalance:* We first analyze whether MOI is a practical bottleneck by examining how camera-side corruptions affect LiDAR-camera fusion models.

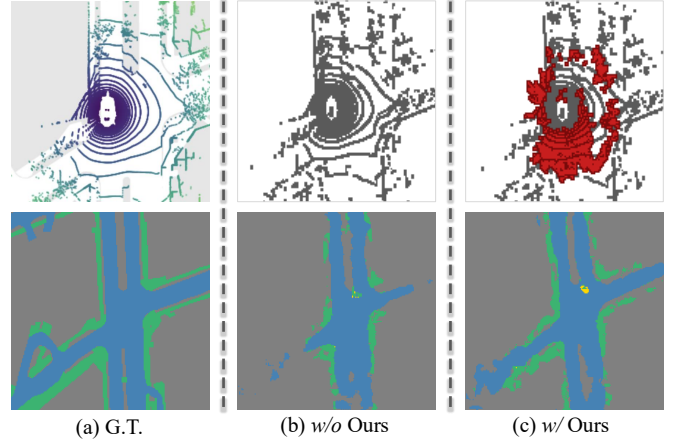


Fig. 4. Impact of Mo-Bal. Sampled image features are shown at the top, and the resulting BEV class predictions are shown at the bottom. Additional samples generated by Mo-Bal improve BEV coverage and lead to more complete semantic maps.

On nuScenes-C [46], three corruption types are applied only to camera inputs across five severity levels, while LiDAR inputs remain unchanged. As shown in Table II, DB-Fusion

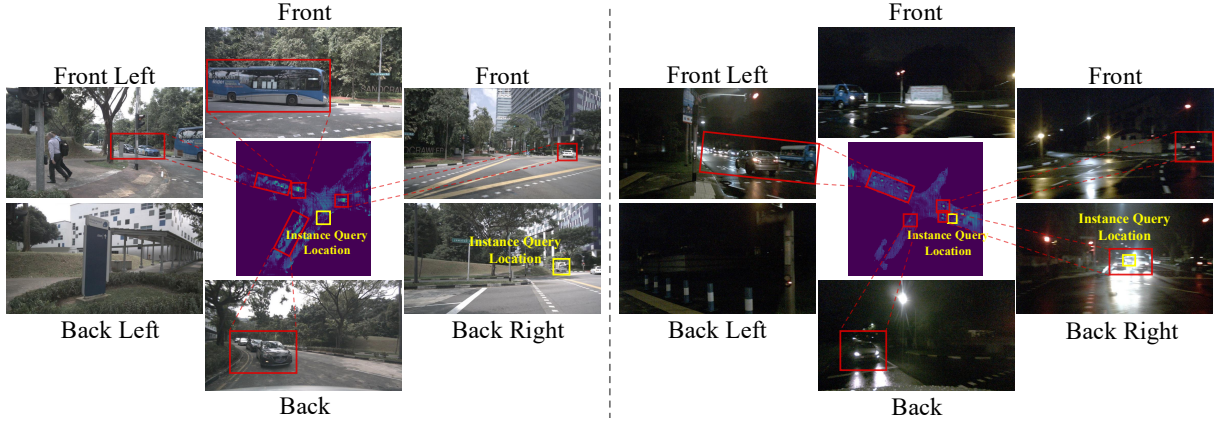


Fig. 5. Visualization of semantic-guided attention maps. The instance feature highlighted with a yellow box serves as the query and aggregates information from BEV regions belonging to the same semantic class, which are highlighted with red boxes.

TABLE IV
ABLATION STUDY ACROSS DUAL-BALANCED DESIGN, MODALITY-OCCUPANCY IMBALANCE, AND CONTEXT QUERY COUNT.

MOI	SOI	mAP	NDS	Components	mAP	NDS	# Query	mAP	NDS
		70.7	72.6	Baseline-LC	70.7	72.6	16	72.3	73.6
✓		72.3	73.8	+ Mo-Bal	71.7	73.3	32	72.5	74.0
	✓	72.8	74.1	+ GeoFuse	71.4	73.1	64	73.2	74.5
✓	✓	73.2	74.5	+ Mo-Bal w/ GeoFuse	72.3	73.8	128	70.2	72.5

(a) Dual Balance

(b) Modality Imbalance

(c) # Context Query

consistently achieves the highest mAP across all corruption types and severity levels, and obtains the best average mAP and NDS for all three corruption types. It outperforms IS-Fusion [8] by up to 2.56 mAP and MambaFusion [10] by up to 1.87 mAP under Gaussian noise. IS-Fusion [8] samples image features only at LiDAR locations, which limits the utilization of camera information, whereas MambaFusion [10] lifts dense image features into BEV, which maximizes the use of camera features. The fact that both extremes degrade more than DB-Fusion under camera corruptions indicates that neither discarding nor fully relying on dense image features is sufficient. Instead, robust fusion requires balancing the two modalities so that neither dominates the BEV representation.

2) *Analysis of Semantic-Occupancy Imbalance*: We next examine whether SOI leads to missed detections by analyzing false-negative (FN) counts. Table III(a) reports FN counts under varying center-distance matching thresholds τ . DB-Fusion consistently achieves the lowest FN count at every threshold, and the gap over MambaFusion increases from 242 at $\tau = 0.5$ m to 572 at $\tau = 4.0$ m. This widening gap indicates that the advantage of DB-Fusion extends beyond tightly localized objects to loosely matched objects, where background interference can become a major cause of missed detections.

Table III(b) further breaks down FN counts by range. The reduction over IS-Fusion increases from 50 at close range, around 10 m, to 172 at long range, around 50 m. This confirms that class-specific context becomes increasingly important as raw physical signals become weaker with distance. These results validate that SOI is a practical source of missed detections and that DB-Fusion effectively mitigates its impact.

TABLE V
ABLATION OF SEMANTIC-BALANCED ENHANCEMENT.

Method	mAP	NDS
Baseline (Mo-Bal + GeoFuse)	72.3	73.8
+ Semantic-Guided Context Attention	72.9	74.3
+ Semantic-Guided Instance Attention	73.2	74.5

D. Ablation Studies

1) *Dual-Balance Ablation*: Table IV(a) isolates the contributions of the MOI and SOI components. Starting from the 70.7 mAP baseline, mitigating MOI alone improves the performance by 1.6 mAP, while mitigating SOI alone provides a larger improvement of 2.1 mAP. This indicates that semantic-occupancy imbalance is the more severe bottleneck in this setting. Combining both components achieves 73.2 mAP and 74.5 NDS, adding 0.4 mAP over the better single component. This confirms that MOI and SOI interact with each other and that joint mitigation provides complementary gains.

2) *Impact of MOI Mitigation*: Table IV(b) breaks down the MOI component. Mo-Bal alone improves mAP by 1.0 by enriching sparse regions with depth-guided pseudo-points, while GeoFuse alone improves mAP by 0.7 by weighting modalities according to geometric reliability. Their combination yields 72.3 mAP and 73.8 NDS, showing a synergistic effect between balanced sampling and geometric reliability estimation. This suggests that Mo-Bal provides GeoFuse with better-distributed features to weight, resulting in more effective cross-modal fusion.

Fig. 4 visualizes this effect. The pseudo-points generated

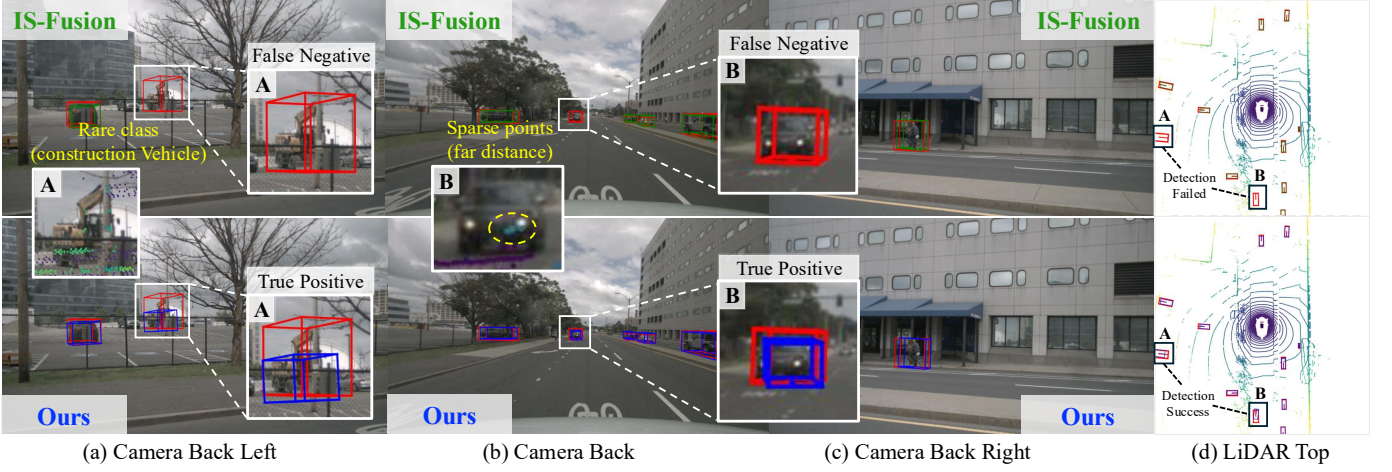


Fig. 6. Qualitative detection results. Red boxes denote ground-truth annotations, green boxes denote IS-Fusion predictions [8], and blue boxes denote DB-Fusion predictions. The examples show that DB-Fusion reduces false negatives even when LiDAR points are sparse or the target object belongs to a rare class.

by Mo-Bal fill coverage gaps in the BEV map, producing more complete semantic predictions. These enhanced semantic predictions further benefit the class-specific partitioning in Se-Bal.

3) *Impact of SOI Mitigation*: We analyze Se-Bal from three aspects. First, Table IV(c) sweeps the number of context queries. Using 64 queries achieves the best balance, with 73.2 mAP and 74.5 NDS. Increasing the number of queries to 128 degrades the performance to 70.2 mAP and 72.5 NDS, suggesting that overly large query banks may introduce redundant context and interfere with class-specific aggregation.

Second, Table V decomposes Se-Bal into its two stages. Semantic-guided context attention improves mAP by 0.6 by routing context within semantic classes, and semantic-guided instance attention adds a further 0.3 mAP by refining instance-level features through same-class self-attention. Third, Fig. 5 visualizes the instance-query attention maps. The instance feature in the highlighted box acts as the query and gathers information from BEV regions of the same semantic class, confirming that Se-Bal learns semantically coherent representations rather than attending indiscriminately to the entire scene.

E. Qualitative Analysis

Fig. 6 compares DB-Fusion with IS-Fusion [8] on challenging detection cases. In Fig. 6(a), DB-Fusion correctly detects a rare-class construction vehicle that IS-Fusion misses, showing that Se-Bal’s class-specific context aggregation can recover weak foreground signals under SOI. In Fig. 6(b), DB-Fusion detects an object at long range despite severe LiDAR sparsity, demonstrating that Mo-Bal and GeoFuse maintain stable object evidence under MOI. Both cases correspond to false negatives in the baseline, which is consistent with our thesis that the two imbalances are primary sources of missed detections.

V. CONCLUSION

In this paper, we presented DB-Fusion, a unified framework for BEV-based LiDAR–camera fusion that jointly mitigates

two structural bottlenecks: Modality-Occupancy Imbalance (MOI) and Semantic-Occupancy Imbalance (SOI). To address MOI, DB-Fusion employs Mo-Bal to compensate for the cross-modal density gap and GeoFuse to adaptively fuse LiDAR and camera features according to geometry-dependent sensor reliability. To address SOI, DB-Fusion introduces Se-Bal, which partitions the BEV representation into semantic classes and performs class-consistent context aggregation and instance-scene interaction. Experiments on nuScenes demonstrate that jointly mitigating MOI and SOI yields strong detection performance, achieving 73.2 mAP on the validation set and 73.6 mAP on the test set with competitive NDS. The ablation studies further show that the largest gain comes from addressing the two imbalances together, rather than treating either one in isolation. In addition, our analysis reveals that DB-Fusion primarily improves detection by reducing false negatives, particularly under sparse point-cloud conditions, while avoiding over-reliance on a single modality. These results indicate that modality density and semantic context should be explicitly balanced in multimodal BEV perception, and suggest a promising direction for reliable 3D object detection in autonomous driving.

REFERENCES

- [1] A. Piazzoni, J. Cherian, J. Dauwels, and L.-P. Chau, “Pem: Perception error model for virtual testing of autonomous vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 1, pp. 670–681, 2023.
- [2] Y. Peng, Y. Qin, X. Tang, Z. Zhang, and L. Deng, “Survey on image and point-cloud fusion-based object detection in autonomous vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 22 772–22 789, 2022.
- [3] X. Wang, K. Li, and A. Chehri, “Multi-sensor fusion technology for 3d object detection in autonomous driving: A review,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 2, pp. 1148–1165, 2023.
- [4] Y. Cui, R. Chen, W. Chu, L. Chen, D. Tian, Y. Li, and D. Cao, “Deep learning for image and point cloud fusion in autonomous driving: A review,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 722–739, 2021.
- [5] Z. Liu, H. Tang, A. Amini, X. Yang, H. Mao, D. Rus, and S. Han, “Bevfusion: Multi-task multi-sensor fusion with unified bird’s-eye view representation,” *arXiv preprint arXiv:2205.13542*, 2022.

- [6] T. Liang, H. Xie, K. Yu, Z. Xia, Z. Lin, Y. Wang, T. Tang, B. Wang, and Z. Tang, "Bevfusion: A simple and robust lidar-camera fusion framework," *Advances in Neural Information Processing Systems*, vol. 35, pp. 10421–10434, 2022.
- [7] J. Philion and S. Fidler, "Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d," in *European conference on computer vision*. Springer, 2020, pp. 194–210.
- [8] J. Yin, J. Shen, R. Chen, W. Li, R. Yang, P. Frossard, and W. Wang, "Is-fusion: Instance-scene collaborative fusion for multimodal 3d object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 14905–14915.
- [9] X. Bai, Z. Hu, X. Zhu, Q. Huang, Y. Chen, H. Fu, and C.-L. Tai, "Transfusion: Robust lidar-camera fusion for 3d object detection with transformers," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 1090–1099.
- [10] H. Wang, J. Gao, W. Hu, and Z. Zhang, "Mambafusion: Height-fidelity dense global fusion for multi-modal 3d object detection," *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2025.
- [11] X. Chen, H. Ma, J. Wan, B. Li, and T. Xia, "Multi-view 3d object detection network for autonomous driving," in *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, 2017, pp. 1907–1915.
- [12] S. Vora, A. H. Lang, B. Helou, and O. Beijbom, "Pointpainting: Sequential fusion for 3d object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 4604–4612.
- [13] J. Huang, Y. Ye, Z. Liang, Y. Shan, and D. Du, "Detecting as labeling: Rethinking lidar-camera fusion in 3d object detection," in *European Conference on Computer Vision*. Springer, 2024, pp. 439–455.
- [14] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuscenes: A multimodal dataset for autonomous driving," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 11621–11631.
- [15] Y. Zhou and O. Tuzel, "Voxelnet: End-to-end learning for point cloud based 3d object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 4490–4499.
- [16] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "Pointpillars: Fast encoders for object detection from point clouds," in *Proceedings of the IEEE/CVF conference on computer vision*, 2019, pp. 12697–12705.
- [17] J. Huang, G. Huang, Z. Zhu, Y. Ye, and D. Du, "Bevdet: High-performance multi-camera 3d object detection in bird-eye-view," *arXiv preprint arXiv:2112.11790*, 2021.
- [18] Y. Li, Z. Ge, G. Yu, J. Yang, Z. Wang, Y. Shi, J. Sun, and Z. Li, "Bevdepth: Acquisition of reliable depth for multi-view 3d object detection," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, no. 2, 2023, pp. 1477–1485.
- [19] C. Lin, D. Tian, X. Duan, J. Zhou, D. Zhao, and D. Cao, "Cl3d: Camera-lidar 3d object detection with point feature enhancement and point-guided fusion," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 18040–18050, 2022.
- [20] M. Liu, Y. Chen, J. Xie, Y. Zhu, Y. Zhang, L. Yao, Z. Bing, G. Zhuang, K. Huang, and J. T. Zhou, "Menet: Multi-modal mapping enhancement network for 3d object detection in autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9397–9410, 2024.
- [21] C.-H. Wang, H.-W. Chen, Y. Chen, P.-Y. Hsiao, and L.-C. Fu, "Vopifnet: Voxel-pixel fusion network for multi-class 3d object detection," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 8527–8537, 2024.
- [22] Y. Li, Y. Yang, and Z. Lei, "Corenet: Conflict resolution network for point-pixel misalignment and sub-task suppression of 3d lidar-camera object detection," *Information Fusion*, vol. 118, p. 102896, 2025.
- [23] Y. Xie, C. Xu, M.-J. Rakotosaona, P. Rim, F. Tombari, K. Keutzer, M. Tomizuka, and W. Zhan, "Sparsefusion: Fusing multi-modal sparse representations for multi-sensor 3d object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 17591–17602.
- [24] H. Wang, H. Tang, S. Shi, A. Li, Z. Li, B. Schiele, and L. Wang, "Unitr: A unified and efficient multi-modal transformer for bird's-eye-view representation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 6792–6802.
- [25] H. Zhang, L. Liang, P. Zeng, X. Song, and Z. Wang, "Sparselif: High-performance sparse lidar-camera fusion for 3d object detection," in *European Conference on Computer Vision*. Springer, 2024, pp. 109–128.
- [26] M. Chen, M. Yang, Y. Zhang, T. Han, X. Li, H. Zhao, and P. Liu, "Multi-modal bev enhancement fusion for 3d object detection in autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, 2025.
- [27] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10012–10022.
- [28] Z. Yao, J. Ai, B. Li, and C. Zhang, "Efficient detr: improving end-to-end object detector with dense prior," *arXiv preprint arXiv:2104.01318*, 2021.
- [29] H. W. Kuhn, "The hungarian method for the assignment problem," *Naval research logistics quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.
- [30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980–2988.
- [31] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.
- [32] Y. Yan, Y. Mao, and B. Li, "Second: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, 2018.
- [33] J. Choe, H. Cho, and Y. Chung, "Performance verification of autonomous driving lidar sensors under rainfall conditions in darkroom," *Sensors*, vol. 24, no. 1, p. 14, 2023.
- [34] T. Yang, W. Yan, J. Lai, Y. Zhao, Z. Wu, Y. Ji, C. Wang, and Z. Li, "Ranging accuracy difference correction among channels of a vehicle-borne road detection lidar," *Optics & Laser Technology*, vol. 174, p. 110477, 2024.
- [35] H. Hu, Z. Liu, S. Chitlangia, A. Agnihotri, and D. Zhao, "Investigating the impact of multi-lidar placement on object detection for autonomous driving," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 2550–2559.
- [36] C. Wang, C. Ma, M. Zhu, and X. Yang, "Pointaugmenting: Cross-modal augmentation for 3d object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 11794–11803.
- [37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [38] T. Yin, X. Zhou, and P. Krähénbühl, "Multimodal virtual point 3d detection," *Advances in Neural Information Processing Systems*, vol. 34, pp. 16494–16507, 2021.
- [39] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 2403–2412.
- [40] T. Liang, X. Chu, Y. Liu, Y. Wang, Z. Tang, W. Chu, J. Chen, and H. Ling, "Cbnet: A composite backbone network architecture for object detection," *IEEE Transactions on Image Processing*, vol. 31, pp. 6893–6906, 2022.
- [41] Z. Yang, J. Chen, Z. Miao, W. Li, X. Zhu, and L. Zhang, "Deepinteraction: 3d object detection via modality interaction," *Advances in Neural Information Processing Systems*, vol. 35, pp. 1992–2005, 2022.
- [42] J. Yan, Y. Liu, J. Sun, F. Jia, S. Li, T. Wang, and X. Zhang, "Cross modal transformer: Towards fast and robust 3d object detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 18268–18278.
- [43] Q. Cai, Y. Pan, T. Yao, C.-W. Ngo, and T. Mei, "Objectfusion: Multi-modal 3d object detection with object-centric fusion," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 18067–18076.
- [44] Y. Lee and J. Park, "Centermask: Real-time anchor-free instance segmentation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 13906–13915.
- [45] T. Qu, D. Liu, J. Zhao, and H. Chen, "Smdfusion: Multimodal sensor fusion for auto driving perception under mixed-traffic scenarios," *IEEE Transactions on Intelligent Transportation Systems*, 2026.
- [46] Y. Dong, C. Kang, J. Zhang, Z. Zhu, Y. Wang, X. Yang, H. Su, X. Wei, and J. Zhu, "Benchmarking robustness of 3d object detection to common corruptions," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 1022–1032.